

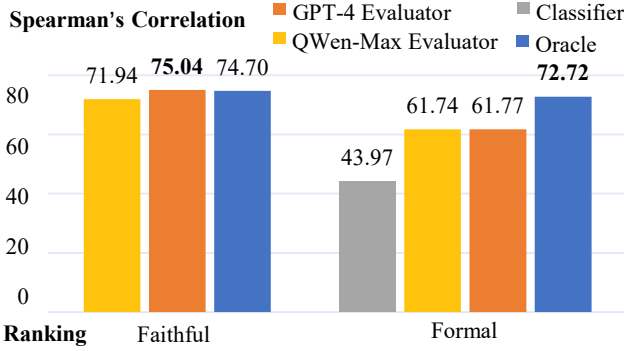


Figure 3: The ranking correlation coefficients between different evaluation methods and human evaluation. Note that the “Oracle” represents the average scores of three annotators, and “Classifier” only reports results on English paragraphs.

Discussion on LLM Evaluation

To measure the reliability of LLM evaluation, we compare the results with human evaluation. Consequently, we find that **the LLM Evaluator is reliable for both faithfulness and formality evaluation**. Based on more than 200 sampled paragraphs, we engage three annotators to independently evaluate and rank the performance of GPT-4, QWen-Max, and QWen-14B in terms of faithfulness and formality. Then we calculate the Spearman’s correlation coefficients (Spearman 1961) between the rankings given by the human evaluation and the LLM evaluation, as shown in Figure 3.

Faithfulness Score: The Spearman’s correlation coefficient of the faithfulness rankings between the LLM Evaluator and human evaluation is 0.75, which is comparable to human performance. This demonstrates the reliability of the LLM Evaluator in faithfulness scoring. Additionally, this indicates that the LLM performs well in determining whether the results faithfully preserve the original content.

Formality Score: As shown in Figure 3, the Spearman’s correlation coefficient between the LLM evaluation and human evaluation is 0.61. Although there is a gap compared to human performance, there is also a strong correlation, which validates that the formality scores are reliable. Moreover, we compare the commonly used evaluation method. Using a RoBERTa-based classifier¹⁰, we classify English paragraphs as formal or informal and assign the formality probability as the formality score. The LLM evaluator outperforms the classifier in the correlation coefficient, confirming the reliability of LLM evaluation.

As Figure 3 illustrates, GPT-4 Evaluator correlates more strongly with human evaluations than QWen-Max Evaluator, confirming its reliability. In addition, we evaluate GPT-4 results three times. The average standard deviations of faithfulness scores (0.19) and formality scores (0.11) across all samples confirm the robustness. Despite LLM updates rapidly, stable evaluation results can still be achieved with the same model version at a low temperature.

Distribution of Languages and Domains

As indicated in Table 6, LLMs generally excel more in Chinese than English at the chunk level. There are more blank

Model	BLEURT↑		S-Faithful↑	
	Chinese	English	Chinese	English
GPT-4	66.95	55.52	7.83	6.80
QWen-Max	57.67	40.79	6.99	5.00
QWen-14B	36.73	21.05	4.77	2.63
LLaMA-8B	36.55	46.03	7.33	5.36

Table 6: The performance of LLMs across different languages.

Model	CK-Recall↑		
	Podcast	Meeting	Lecture
GPT-4 (chunk)	19.82	30.00	32.81
w/ aux. (Origin)	45.82	30.00	36.72
w/ aux. (RAG)	36.55	35.00	39.84
w/ aux. (Summary)	41.27	35.00	37.50
w/ aux. (Keywords)	47.27	45.00	37.50

Table 7: The CK-Recall performance of GPT-4 with various auxiliary information methods across different domains.

paragraphs in English (2.68% for GPT-4) than in Chinese (0.83%). This may be because LLMs prioritize fluency in English over Chinese, often merging paragraphs for better fluency at the expense of faithfulness.

As shown in Table 7, **the podcast domain, with the longest auxiliary information, shows the most significant improvement in CK-Recall**, especially with the Keywords method. This indicates that plentiful auxiliary information aids ASR error correction improvement. Optimizing the use of long-text auxiliary information for further improvement is an area that warrants deeper exploration.

Conclusion

To improve the readability of ASR transcripts, we propose the Contextualized Spoken-to-Written conversion (CoS2W) task, construct and make available the document-level and multi-domain Spoken2Written of ASR transcripts Benchmark (SWAB) dataset. Based on the SWAB dataset, we compare the performance of different LLMs at various granularity levels, verify the beneficial roles of contexts and auxiliary information, and find that it is worth further exploring how LLMs maintain faithfulness and utilize auxiliary information to enhance ASR Error Correction. Compared with human evaluation, we find that the LLM Evaluator performs well in faithfulness and formality ranking with a good correlation. In the future, we plan to expand the dataset scale to better support research on supervised training for this task.

Limitations

The study’s limitation lies in the SWAB dataset having only a single target, which may inadequately reflect model performance through many automatic metrics. Future efforts will focus on offering diverse targets and developing more effective evaluation methods. Note that the source data of SWAB are owned by the copyright holder. We only provide ASR transcripts and annotation targets, along with the corresponding links of audios and videos. The license of SWAB will be for research purposes only. Additionally, it is inevitable that evaluations are influenced by the biases of GPT-4, given that we obtain initial results through GPT-4 and also use it for experiments and evaluation.

¹⁰<https://www.huggingface.co/s-nlp/roberta-base-formality-ranker>

Ethics Statement

The SWAB dataset used in this research is strictly for academic and non-commercial purposes. We implemented several measures to ensure compliance with ethical standards, as follows.

- **Data Transparency and Anonymization.** The Chinese meetings in the SWAB are sourced from the training set of the publicly available AliMeeting dataset, and the English meetings come from the publicly available AMI meeting dataset. For our collected podcasts and lectures, we only provide ASR transcripts after rigorous text anonymization processes and our annotations, to ensure transparency regarding the data sources and their usage while maintaining anonymity.
- **Data Access Compliance.** To further ensure the ethical use of the dataset, we require researchers to contact us via email to confirm their compliance with ethical guidelines and the conditions outlined in our data usage declaration, before granting them access to the dataset. This procedure includes ensuring that they are aware of and adhere to the Personal Information Protection Law (PIPL) and any relevant legal frameworks regarding personal data usage.
- **Authorization.** Any personal data should be used only with express authorization, ensuring lawful and fair processing in accordance with applicable laws.

Acknowledgments

We would like to express gratitude to the anonymous reviewers for their valuable feedback.

References

- Bryant, C.; Felice, M.; Andersen, Ø. E.; and Briscoe, T. 2019. The BEA-2019 Shared Task on Grammatical Error Correction. In Yannakoudakis, H.; Kochmar, E.; Leacock, C.; Madnani, N.; Pilán, I.; and Zesch, T., eds., *BEA@ACL 2019*, 52–75. Association for Computational Linguistics.
- Carletta, J.; Ashby, S.; Bourban, S.; Flynn, M.; Guillemot, M.; Hain, T.; Kadlec, J.; Karaiskos, V.; Kraaij, W.; Kronenthal, M.; et al. 2005. The AMI meeting corpus: A pre-announcement. In *International workshop on machine learning for multimodal interaction*, 28–39. Springer.
- Chen, G.; Chai, S.; Wang, G.; Du, J.; Zhang, W.; Weng, C.; Su, D.; Povey, D.; Trmal, J.; Zhang, J.; Jin, M.; Khudanpur, S.; Watanabe, S.; Zhao, S.; Zou, W.; Li, X.; Yao, X.; Wang, Y.; You, Z.; and Yan, Z. 2021. GigaSpeech: An Evolving, Multi-Domain ASR Corpus with 10, 000 Hours of Transcribed Audio. In Hermansky, H.; Cernocký, H.; Burget, L.; Lamel, L.; Scharenborg, O.; and Motlíček, P., eds., *Inter-speech 2021*, 3670–3674. ISCA.
- Daudaravicius, V.; Banchs, R. E.; Volodina, E.; and Napoles, C. 2016. A Report on the Automatic Evaluation of Scientific Writing Shared Task. In Tetreault, J. R.; Burstein, J.; Leacock, C.; and Yannakoudakis, H., eds., *BEA@NAACL-HLT 2016*, 53–62. The Association for Computer Linguistics.
- Fang, T.; Yang, S.; Lan, K.; Wong, D. F.; Hu, J.; Chao, L. S.; and Zhang, Y. 2023. Is ChatGPT a Highly Fluent Grammatical Error Correction System? A Comprehensive Evaluation. *CoRR*, abs/2304.01746.
- Gao, Z.; Zhang, S.; McLoughlin, I.; and Yan, Z. 2022. Paraformer: Fast and accurate parallel transformer for non-autoregressive end-to-end speech recognition. *arXiv preprint arXiv:2206.08317*.
- Guo, Z.; Yu, L.; Xu, M.; Jin, R.; and Xiong, D. 2023. CS2W: A Chinese Spoken-to-Written Style Conversion Dataset with Multiple Conversion Types. In Bouamor, H.; Pino, J.; and Bali, K., eds., *EMNLP 2023*, 3962–3979. Association for Computational Linguistics.
- Gupta, A.; Xu, J.; Upadhyay, S.; Yang, D.; and Faruqui, M. 2021. Disfl-QA: A benchmark dataset for understanding disfluencies in question answering. *arXiv preprint arXiv:2106.04016*.
- Han, R.; Soldaini, L.; and Moschitti, A. 2021. Modeling Context in Answer Sentence Selection Systems on a Latency Budget. In *EACL 2021*, 3005–3010.
- He, J.; Yang, Z.; and Toda, T. 2023. ed-cec: improving rare word recognition using asr postprocessing based on error detection and context-aware error correction. In *ASRU 2023*, 1–6. IEEE.
- Ihori, M.; Takashima, A.; and Masumura, R. 2020. Parallel Corpus for Japanese Spoken-to-Written Style Conversion. In Calzolari, N.; Béchet, F.; Blache, P.; Choukri, K.; Cieri, C.; Declerck, T.; Goggi, S.; Isahara, H.; Maegaard, B.; Mariani, J.; Mazo, H.; Moreno, A.; Odijk, J.; and Piperidis, S., eds., *LREC 2020*, 6346–6353. European Language Resources Association.
- Lai, H.; Toral, A.; and Nissim, M. 2023. Multidimensional Evaluation for Text Style Transfer Using ChatGPT. *CoRR*, abs/2304.13462.
- Lakomkin, E.; Wu, C.; Fathullah, Y.; Kalinli, O.; Seltzer, M. L.; and Fuegen, C. 2024. End-to-end speech recognition contextualization with large language models. In *ICASSP 2024*, 12406–12410. IEEE.
- Li, Y.; Huang, H.; Ma, S.; Jiang, Y.; Li, Y.; Zhou, F.; Zheng, H.-T.; and Zhou, Q. 2023. On the (in) effectiveness of large language models for chinese text correction. *arXiv preprint arXiv:2307.09007*.
- Li, Y.; Qin, S.; Ye, J.; Ma, S.; Li, Y.; Qin, L.; Hu, X.; Jiang, W.; Zheng, H.; and Yu, P. S. 2024. Rethinking the Roles of Large Language Models in Chinese Grammatical Error Correction. *CoRR*, abs/2402.11420.
- Lin, C.-Y. 2004. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*, 74–81.
- Ma, R.; Qian, M.; Manakul, P.; Gales, M.; and Knill, K. 2023. Can generative large language models perform asr error correction? *arXiv preprint arXiv:2307.04172*.
- Mai, L.; and Carson-Berndsen, J. 2023. Enhancing conversational quality in language learning chatbots: An evaluation of GPT4 for ASR error correction. *arXiv e-prints*, arXiv:2307.

- Min, Z.; and Wang, J. 2023. Exploring the integration of large language models into automatic speech recognition systems: An empirical study. In *ICONIP*, 69–84. Springer.
- Napoles, C.; Sakaguchi, K.; and Tetreault, J. R. 2017. JFLEG: A Fluency Corpus and Benchmark for Grammatical Error Correction. In Lapata, M.; Blunsom, P.; and Koller, A., eds., *EACL 2017*, 229–234. Association for Computational Linguistics.
- Panayotov, V.; Chen, G.; Povey, D.; and Khudanpur, S. 2015. Librispeech: An ASR corpus based on public domain audio books. In *2015 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2015, South Brisbane, Queensland, Australia, April 19-24, 2015*, 5206–5210. IEEE.
- Papineni, K.; Roukos, S.; Ward, T.; and Zhu, W. 2002. Bleu: a Method for Automatic Evaluation of Machine Translation. In *ACL 2002*, 311–318. ACL.
- Pearson, K. 1895. VII. Note on regression and inheritance in the case of two parents. *proceedings of the royal society of London*, 58(347-352): 240–242.
- Rao, G.; Yang, E.; and Zhang, B. 2020. Overview of NLPTEA-2020 shared task for Chinese grammatical error diagnosis. In *NLPTEA 2020*, 25–35.
- Reif, E.; Ippolito, D.; Yuan, A.; Coenen, A.; Callison-Burch, C.; and Wei, J. 2022. A Recipe for Arbitrary Text Style Transfer with Large Language Models. In Muresan, S.; Nakov, P.; and Villavicencio, A., eds., *ACL 2022*, 837–848. Association for Computational Linguistics.
- Robertson, S. E.; and Zaragoza, H. 2009. The Probabilistic Relevance Framework: BM25 and Beyond. *Found. Trends Inf. Retr.*, 3(4): 333–389.
- Sellam, T.; Das, D.; and Parikh, A. P. 2020. BLEURT: Learning Robust Metrics for Text Generation. In Jurafsky, D.; Chai, J.; Schluter, N.; and Tetreault, J. R., eds., *ACL 2020*, 7881–7892. Association for Computational Linguistics.
- Spearman, C. 1961. The proof and measurement of association between two things.
- Tao, Z.; Xi, D.; Li, Z.; Tang, L.; and Xu, W. 2024. CAT-LLM: Prompting Large Language Models with Text Style Definition for Chinese Article-style Transfer. *CoRR*, abs/2401.05707.
- Wang, Y.; Kong, C.; Yang, L.; Wang, Y.; Lu, X.; Hu, R.; He, S.; Liu, Z.; Chen, Y.; Yang, E.; and Sun, M. 2021. YACL: A Chinese Learner Corpus with Multidimensional Annotation. *CoRR*, abs/2112.15043.
- Yang, C.-H. H.; Gu, Y.; Liu, Y.-C.; Ghosh, S.; Bulyko, I.; and Stolcke, A. 2023. Generative speech recognition error correction with large language models and task-activating prompting. In *ASRU 2023*, 1–8. IEEE.
- Yannakoudakis, H.; Briscoe, T.; and Medlock, B. 2011. A New Dataset and Method for Automatically Grading ESOL Texts. In Lin, D.; Matsumoto, Y.; and Mihalcea, R., eds., *ACL-HLT 2011*, 180–189. The Association for Computer Linguistics.
- Yu, F.; Zhang, S.; Fu, Y.; Xie, L.; Zheng, S.; Du, Z.; Huang, W.; Guo, P.; Yan, Z.; Ma, B.; Xu, X.; and Bu, H. 2022. M2MeT: The ICASSP 2022 Multi-Channel Multi-Party Meeting Transcription Challenge. In *Proc. ICASSP*. IEEE.
- Zhang, B.; Lv, H.; Guo, P.; Shao, Q.; Yang, C.; Xie, L.; Xu, X.; Bu, H.; Chen, X.; Zeng, C.; Wu, D.; and Peng, Z. 2022a. WENETSPEECH: A 10000+ Hours Multi-Domain Mandarin Corpus for Speech Recognition. In *ICASSP 2022*, 6182–6186. IEEE.
- Zhang, Q.; Chen, Q.; Li, Y.; Liu, J.; and Wang, W. 2021a. Sequence model with self-adaptive sliding window for efficient spoken document segmentation. In *ASRU 2021*, 411–418. IEEE.
- Zhang, R.; Wang, X.; Zhang, C.; He, Z.; Wu, H.; Li, Z.; Wang, H.; Chen, Y.; and Li, Q. 2021b. Bstc: A large-scale chinese-english speech translation dataset. *arXiv preprint arXiv:2104.03575*.
- Zhang, X.; Zhang, X.; Yang, C.; Yan, H.; and Qiu, X. 2023. Does Correction Remain A Problem For Large Language Models? *CoRR*, abs/2308.01776.
- Zhang, Y.; Li, Z.; Bao, Z.; Li, J.; Zhang, B.; Li, C.; Huang, F.; and Zhang, M. 2022b. MuCGEC: a Multi-Reference Multi-Source Evaluation Dataset for Chinese Grammatical Error Correction. In Carpuat, M.; de Marneffe, M.; and Ruíz, I. V. M., eds., *NAACL-HLT 2022*, 3118–3130. Association for Computational Linguistics.
- Zhao, Y.; Jiang, N.; Sun, W.; and Wan, X. 2018. Overview of the NLPCC 2018 Shared Task: Grammatical Error Correction. In Zhang, M.; Ng, V.; Zhao, D.; Li, S.; and Zan, H., eds., *NLPCC 2018*, volume 11109 of *Lecture Notes in Computer Science*, 439–445. Springer.
- Zheng, S.; Cheng, L.; Chen, Y.; Wang, H.; and Chen, Q. 2023. 3D-Speaker: A Large-Scale Multi-Device, Multi-Distance, and Multi-Dialect Corpus for Speech Representation Disentanglement. *arXiv preprint arXiv:2306.15354*.